

Haydeé Contreras Peruyero

Postdoctoral Research at
Centro de Ciencias
Matemáticas UNAM



18 Aug 1989



CCM UNAM



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<https://haydeeperuyero.github.io/>

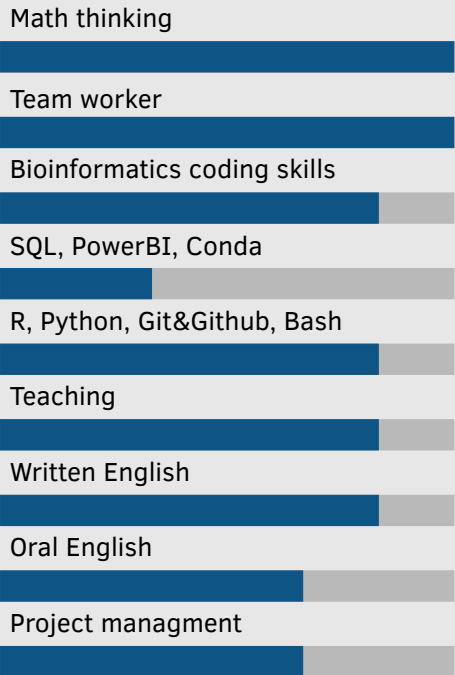


haydeeperuyero@matmor.unam.mx

About me

My research interests lie at the intersection of geometry, topology, and bioinformatics. My work has focused on geometric group theory and the application of Topological Data Analysis (TDA) to problems in genomics, metagenomics, and pangenomics. I also use machine learning and statistics to explore structures in biological data. I have a strong passion for teaching, programming, and learning languages.

Skills



(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Interests

Topological ang Geometric Data Analysis | Bioinformatics | Geometric GroupTheory | Global Riemannian Geometry | Quantum Geometry and Topology

Education

2016-2021	Ph. D. in Science in Mathematics	Instituto de Matemáticas, UNAM
	Thesis: <i>Quasi-Isometric Rigidity in Higher Graph Manifolds</i>	
2020-2021	Specialty in Applied Statistics	Instituto de Investigaciones en Matemáticas Aplicadas y en Sistemas
2014-2016	M.Sc. in Mathematics	Instituto de Matemáticas, UNAM
	Thesis: On the Geometry and Topology of Graph Manifolds	
2007-2012	B.Sc. in Mathematics	Universidad Veracruzana
	Thesis: Triangulations of Hyperbolic 3-Manifolds Fibered over the Circle with Torus with a Hole as Fiber	

Professional Experience

2025-	Postdoctoral Research	CONAHCYTH, CCM UNAM
	<i>Topological Data Analysis Applied to Genomic Data and Cryptography with Geometric Group Theory</i>	
2023-2025	Postdoctoral research DGAPA	CCM UNAM
	Applications of Topological Data Analysis and Quasi-Isometry Invariants	
2022-2023	Postdoctoral research CONACyT Ciencia de Frontera	CONACyT, CCM UNAM
	Invariant Properties of Higher Graph Manifolds	
2021-2022	Head of the Department of Basic Sciences	ITSMT

Recent Awards

2023	1 Price. Critical Assessment of Massive Data Analysis Microbiome Challenge. CAMDA 2023
2022	Mentee in the Mentoring in Action - Women and Girls in STEM Online programme
2022	2nd place at the ISCB-La SOLBIO BIONETMX WIKIPEDIA COMPETITION, Article: "Matrices de distancias"

Publications

2025	Topological data analysis captures Horizontal Gene Transfer in Antimicrobial Resistance gene families among clinically relevant bacteria. <i>Front. Microbiol.</i> 16:1461293. https://doi.org/10.3389/fmicb.2025.1461293
2025	Asymptotic dimension and geometric decompositions in dimensions 3 and 4, H. Contreras Peruyero and P. Suárez-Serrato. <i>Journal of the Australian Mathematical Society</i> . Published online 2025:1-26. https://doi.org/10.1017/S1446788725000072
2024	Multidisciplinary Workshop on Pangenomics and Topological Data Analysis, H. Contreras-Peruyero et al. <i>Bioinformatics Advances</i> , 2024;, vbae139 https://doi.org/10.1093/bioadv/vbae139
2024	CAMDA 2023: finding patterns in urban microbiomes. H. Contreras-Peruyero et al. <i>Front. Genet.</i> (2024) 15:1449461. https://doi.org/10.3389/fgene.2024.1449461
2024	Physicochemical Properties of Two Mexican Stingless Bee Honeys to Strengthen Their Biocultural Value. A. Xolalpa-Aroche et al. (2024). <i>Revista Brasileira de Farmacognosia</i> , 1-11. https://doi.org/10.1007/s43450-024-00566-z

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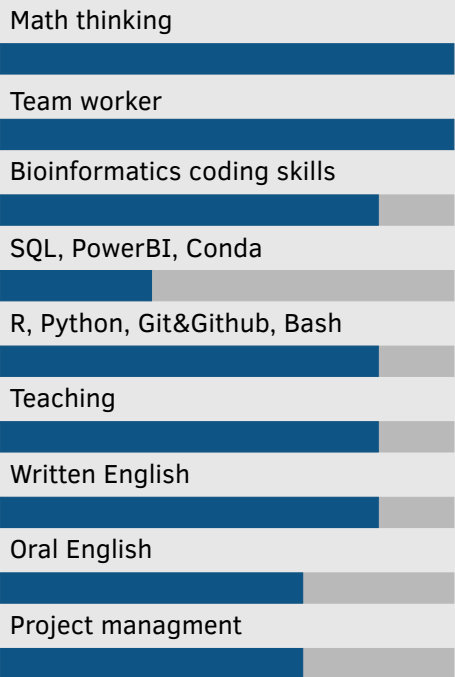
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2024

Actinobacteria bioprospecting from ore-forming environments. C. Aguilar et al. Microbial Genomics, 10(5), 001253. ISSN 2057-5858, <https://doi.org/10.1099/mgen.0.001253>

2022

Collapsing and group growth as obstructions to Einstein metrics on some smooth 4-manifolds. H. Contreras Peruyero adn P. Suárez-Serrato, New York Journal of Mathematics 28 (2022), 659–671. <http://nyjm.albany.edu/j/2022/28-27.html>

Accepted papers

2025

Fractal dimensions of complex networks: advocating for a topological approach. Accepted May 2025.

2025

Linear bounds for the lengths of geodesics on manifolds with curvature bounded below. Accepted April 2025. <https://arxiv.org/abs/2410.10975>

Preprints

2025

New Acetilactobacillus species is the most abundant microorganism in the microbiome of native stingless bee honeys in Mexico. In preparation.

2024

Lengths of the Orthogonal Geodesic Chords on Riemannian Manifolds. In preparation.

2022

On the quasi-isometric rigidity of chambers and walls in cusp-decomposable manifolds. Submitted 2022. <https://arxiv.org/abs/2010.03736>

Selected talks and posters

2025

CMO Workshop: Mixtures of probability and Geometry Oaxaca, Mexico
Detecting Evolutionary Patterns and Cellular Organization through Persistent Homology

2025

La Matemática Atenagorica Michoacan, Mexico
Diversidad microbiana en mieles de abejas nativas sin aguijón de la Península de Yucatán

2024

1st Asia&Pacific Bioinformatics Joint Conference Okinawa, Japan
Poster “Microbial Diversity and Characterization of Stingless Bee Honey from the Yucatán Peninsula”

2024

Escuela de Matemáticas de América Latina y el Caribe Cuernavaca 2024 Morelos, Mexico
Talk “Analizando datos con topología y geometría”

2024

2024 Joint Mathematics Meetings San Francisco, USA
Talk “Asymptotic dimension and geometric decompositions in dimension 4”

2023

XXIV Escuela de Otoño (EOBM) y el XVII Encuentro Nacional (ENBM), of Bio-mathematics Veracruz, Mexico
Minicourse “Aplicaciones de análisis topológico de datos a Pangenómica”

2023

Iberoamerican and Pan Pacific International Conference on Topology and its Applications Puebla, Mexico
“Invariants under quasi-isometries”

2023

31st Conference on Intelligent Systems for Molecular Biology and 22th European Conference On Computational Biology as part of the Annual International Conference on Critical Assessment of Massive Data Analysis Lyon, France
Poster “Topology, regression and metapangenome enrich conventional microbiome analyses”

2023

Workshop Técnicas Modernas en Geometría Riemanniana 2023Mexico City, Mexico
Talk: “Algunos invariantes de cuasi-isometrías”

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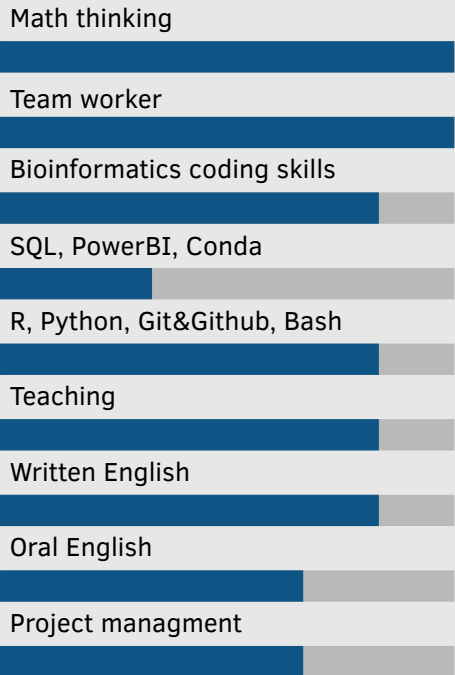


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2023	2023 Joint Mathematics Meetings	Boston, USA
	Talk “Collapsing and group growth as an obstruction to Einstein metrics on some smooth 4-manifolds”	
2022	8th Mexican Workshop on Applied Geometry and Topology	Virtual
	Poster “Topological and Geometric explorations of Autism fMRI connectomes”	

Science communication

2025	Feria de las Matemáticas Workshop “Papiroflexia Modular”	Michoacan, Mexico
2024	Futuras STEM Michoacán Workshop “Matemagia”	Michoacan, Mexico
2024	Feria Infantil y Juvenil de Ciencias y Artes Workshop “Estereogramas”	Michoacan, Mexico
2024	Feria de las Matemáticas Talk “¿Qué hacemos cuentas? Son puros cuentos”	Michoacan, Mexico
2023	11 Fiesta de las Ciencias y Humanidades Workshop “Lengua de Señas Mexicana (LSM) y Matemáticas”	Michoacan, Mexico
2023	11 Fiesta de las Ciencias y Humanidades Workshop “Biomatemáticas”	Michoacan, Mexico

Synergistic Activities

2025	Segundo Encuentro de Mujeres en Topología, Geometría y sus Aplicaciones	Michoacan, Mexico
	Co-organizer, Webpage: https://matmor.unam.mx/encuentro-mujeres-topogeo-2025/	
2025	IV Escuela de Teoría Geométrica de Grupos	Michoacan, Mexico
	Co-organizer, webpage https://matmor.unam.mx/etgg-2025/	
2024	Hackaton Reto Critical Assessment of Massive Data	Michoacan, Mexico
	Co-organizer	
2022-	Long-distance Seminar on Geometric Group Theory in Mexico	Virtual
	Co-organizer	
2022-	R Ladies Morelia	Hybrid, Michoacan, Mexico
	Co-organizer, webpage https://www.meetup.com/es-ES/rladies-morelia/?eventOrigin=home_groups_you_organize	

Formation of Human Resources

2024-	Master’s Thesis Advisor Johanna Atenea Carreón Baltazar	CCM UNAM
	Thesis title “Analysis of the Distribution of Duplicated Genes in Cyanobacteria”	
2024-	Master’s Thesis Co-advisor Luis Raul Figueroa Martinez	CCM UNAM
	Thesis title “Topological Data Analysis for Automated Classification of Multicellular Patterns”	
2024	Graduate course “Advanced Statistics Course: Machine Learning and Deep Geometric Learning”	CCM UNAM
	https://github.com/CursosPosgradoCCM/ML-GDL	
2024	Graduate course “Scientific Computing: Scientific computing for data analysis”	CCM UNAM
	https://haydeeperuyero.github.io/Computo_Cientifico/	
2023	Graduate course “Topology advanced course: Applications of Topological Data Analysis”	CCM UNAM
	https://github.com/HaydeePeruyero/CursoA_Aplicaciones_TDA	
2023	Graduate course “Statistics seminar: Data Analysis with R”	CCM UNAM
	https://haydeeperuyero.github.io/Seminario_Estadistica/	

